A Zero-Parameter Framework for Cosmological Dynamics and Galaxy Cluster Mass Discrepancies via Structural Self-Energy Expansion (SSEE-V3.6)

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Abstract

We present the Structural Self-Energy Expansion (SSEE-V3.6), a minimalist theoretical framework that unifies cosmological expansion and galaxy-cluster dynamics using only two dimensionless geometric constants — π and the golden ratio φ — without introducing free parameters or exotic dark-matter particles. From these two axioms, a closed set of seven structural registers is derived algebraically, culminating in the dark-energy equation of state

$$w_0 = -\frac{3\varphi + \pi}{2(\varphi + \pi)} \approx -0.840, \qquad w_a = -\frac{2\varphi + \pi}{2(\varphi + \pi)} \approx -0.670.$$

No fitting procedure is involved: w_0 and w_a are uniquely fixed by the geometry of φ and π. The asymptotic structural viscosity factor $\mathrm{KAL}_0 = \beta + \pi \approx 5.5214$ natively reproduces the observed cosmic dark-to-baryonic matter density ratio. At galactic scales, SSEE derives an acceleration threshold $a_0 = c\,H_0/\mathrm{KAL}_0 \approx 1.186 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, consistent with the empirical MOND scale. Incorporating a light-neutrino mass sum $\sum m_\nu = 0.085\,\mathrm{eV}$, the framework structurally resolves the megaparsec cluster mass discrepancy when combined with an IGIMF-corrected baryonic baseline [Zhang et al., 2026]. The Hubble tension is addressed through a structural local correction: the dimensionless viscosity coupling $\mathrm{KAL}_0 \approx 5.521$ reproduces the observed offset $H_{0,\mathrm{local}} - H_{0,\mathrm{global}} \approx 5.7\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ at the $\sim 3\%$ level. The dimensional completion of this coincidence — deriving the velocity-per-length scale from the IS viscosity equations — is reserved for a dedicated paper. A full Bayesian MCMC validation against DESI DR2 [Abdul-Karim et al., 2025], Planck 2018, and galaxy-cluster mass data ($\Delta\mathrm{BIC} = -5.55$ for the dynamic sector relative to ΛCDM) is presented in the companion Paper 2 [Almeida Vallejo, 2026]. The framework is falsifiable: a B-mode polarisation tensor-to-scalar ratio $r = \varphi^{-10} = 0.00813$ is predicted for LiteBIRD ($\sim$2032).

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1 Introduction and Theoretical Foundations

The SSEE framework operates under the principle of algebraic closure and macroscopic structural self-energy. Rather than adding particle degrees of freedom, the system's identity is defined by a closed set of geometric invariants derived exclusively from π and Φ. These fundamental scales dictate the macroscopic behaviour of spacetime:

- $\boldsymbol{\Omega}$ ($\pi + \Phi \approx 4.7596$): Stability Metric.
- $\boldsymbol{\beta}$ ($(\pi + \Phi)/2 \approx 2.3798$): Base Coupling Scalar.
- $\mathbf{KAL_0}$ ($\beta + \pi \approx 5.5214$): Asymptotic Structural Viscosity.
- $\boldsymbol{P}$ ($\Omega + \Phi \approx 6.3776$): Dynamical Evolution Scalar.
- $\boldsymbol{K_v}$ ($\Phi + \pi + \Omega \approx 9.5192$): Structural Constraint Invariant.
- $\boldsymbol{T_r}$ ($3(\Phi + \beta) \approx 11.9935$): 3D Saturation Horizon.
- $\boldsymbol{M_v}$ ($\Phi + \pi + K_v \approx 14.2788$): Maximal Dimensional Invariant.

1.1 Predictive Register

A common objection to algebraic models is selection bias: that the constant combinations are chosen post-hoc to reproduce known values. Table 1 documents the chronological sequence of the SSEE derivations relative to the observational data they are compared against. The Genesis 5.12 compendium (Zenodo DOI: `10.5281/zenodo.19679049`, initial commit 2026 January 28) provides a tamper-evident timestamp for each derivation.

Table 1: Predictive Register: SSEE derivations in chronological order. "Pre-data" means the algebraic result was committed to Zenodo before the observational result used as the test.

Quantity	Algebraic formula	Test dataset	Status
$w_0 = -T_r/M_v$	-0.8399	DESI DR2 (2025)	Pre-data
$w_a = -P_{sc}/K_v$	-0.6699	DESI DR2 (2025)	Pre-data
MIRA $= (\varphi + \beta)/2$	1.9989	CMB/BAO Ω_m ratio	Pre-data
$\Omega_m = (\pi - \varphi)/(\pi + \varphi)$	0.3201	Planck 2018	Retrodiction
$n_s = 1 - \varphi^{-7}$	0.96556	Planck 2018	Retrodiction
$H_0 = 3(\varphi + \pi)^2$	67.962	Planck 2018	Retrodiction
$r = \varphi^{-10}$	0.00813	LiteBIRD ($\sim$2032)	Prediction

"Retrodiction" denotes results derived independently and then compared against previously published values; no optimisation over the algebraic formula space was performed to match a target value. The distinction between pre-data predictions and retrodictions is explicitly noted to maintain transparency.

1.2 Axiomatic Principles

SSEE is explicitly constructed as a top-down theoretical ansatz. Its internal symmetries are not phenomenological fits derived bottom-up, but fundamental axiomatic identities that the macroscopic universe must obey. Three physical postulates govern the interactions of these geometric invariants:

1. **The Temporal Triad (Cosmological Epochs):** Spacetime evolves through three distinct thermodynamic states — the Origin/Law phase (initial singularity), the Ignition/Processing phase (structure formation), and the Transcendental Saturation phase (dark energy asymptotic domination). The Hubble history $H(a)$ is bounded by:
$$\lim_{a\to 0} H^2(a) \propto \rho_{\rm bar}, \qquad \lim_{a\to\infty} H^2(a) = \frac{8\pi G}{3}\,\rho_{\rm crit,0}\left(\frac{T_r}{M_v}\right). \tag{1}$$
2. **Non-Self-Addition (Evolution by Difference):** Structural evolution proceeds through non-linear geometric scaling rather than linear superposition. All macroscopic observables $\mathcal{O}$ must be strictly functional dependencies of the invariant set and H_0; no additional free parameters are permitted:
$$\mathcal{O} = f(\Omega,\, \beta,\, \mathrm{KAL}_0,\, P,\, K_v,\, T_r,\, M_v;\, H_0). \tag{2}$$
3. **Asymmetric Equivalence:** Energy states can possess identical geometric values but divergent dynamic functions. The macroscopic geometric response must inversely mirror the physical source:
$$\mu_{\rm SSEE}(x)\cdot \mathrm{KAL}(x) \equiv 1. \tag{3}$$

1.3 The SSEE Effective Action

Instead of an arbitrary cosmological constant Λ, the vacuum is governed by a static geometric pressure field tied to the critical density $\rho_{\rm crit,0} = 3H_0^2/(8\pi G)$:

$$S_{\rm SSEE} = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} - \rho_{\rm crit,0}\left(\frac{T_r}{M_v}\right) + \mathcal{L}_{\rm matter}\right]. \tag{4}$$

This action guarantees that the geometric constraints are not phenomenological fits, but inevitable consequences of the universe's macroscopic structural self-energy at saturation.

Falsifiability. The framework is strictly falsifiable. It would be refuted by: (1) direct detection of a collisionless dark-matter particle accounting for the missing mass; (2) definitive cosmological constraints proving $w_a = 0$; or (3) high-significance confirmation of a normal neutrino hierarchy with $\sum m_\nu < 0.06\,\mathrm{eV}$.

2 Effective Field Theory Embedding

2.1 The SSEE Action

The SSEE dark-energy sector is described by a k-essence scalar field ϕ minimally coupled to gravity and extended by a geometric bulk-viscosity term in the Eckart approximation [Eckart, 1940, Weinberg, 1971]:

$$S = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} + P(X) - \zeta\left(\nabla_\mu u^\mu\right)^2 + \mathcal{L}_m\right], \tag{5}$$

where $X \equiv -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$ is the canonical kinetic term. The fluid four-velocity of the dark-energy component is identified with the normalised gradient of the field,

$$u_\mu = -\frac{\partial_\mu\phi}{\sqrt{2X}}, \tag{6}$$

so that $u_\mu u^\mu = -1$ is satisfied automatically, and the viscous and kinetic sectors remain self-consistently defined by the single degree of freedom ϕ.

2.2 Algebraic Determination of $P(X)$

For a power-law k-essence Lagrangian $P(X) = \mathcal{A}\,X^n$ the equation of state is *exactly* constant:

$$w_\phi = \frac{P}{2XP_X - P} = \frac{\mathcal{A}X^n}{(2n-1)\mathcal{A}X^n} = \frac{1}{2n-1}. \tag{7}$$

Imposing the SSEE algebraic constraint $w_0 = -T_r/M_v$ uniquely fixes the kinetic exponent without any fitting:

$$\boxed{n = \frac{T_r - M_v}{2\,T_r} = \frac{1+w_0}{2w_0} \approx -0.09527,} \tag{8}$$

where $T_r = 3(\varphi+\beta)$ and $M_v = \varphi + \pi + K_v$ are the SSEE structural constants defined in Section 1.2. One verifies immediately that $1/(2n-1) = -T_r/M_v = w_0$ identically.

The overall normalisation is fixed by requiring that the present-day dark-energy density equals the SSEE prediction:

$$\rho_{\mathrm{DE},0} = \frac{3H_0^2}{8\pi G}\,\frac{T_r}{M_v}, \tag{9}$$

with H_0 constrained observationally to $66.75^{+0.44}_{-0.44}$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ by the MCMC analysis of Paper 2. Given the energy scale $\rho_{\mathrm{DE},0}$, the amplitude reads

$$\mathcal{A} = \frac{\rho_{\mathrm{DE},0}^{1-n}}{2n-1}, \tag{10}$$

which completes the specification of $P(X)$ with no free parameters beyond H_0.

Remark on the CPL evolution. The pure k-essence sector (7) delivers a *constant* $w = w_0$. The time variation w_a arises from the viscous source (Section 2.3), and is shown there to emerge algebraically from the scale-factor evolution of the viscosity coefficient.

2.3 Geometric Bulk Viscosity and the Emergence of w_a

2.3.1 Constant-coupling regime (w_0)

The retention constant $\mathrm{KAL}_0 \equiv \beta + \pi \approx 5.521$ governs the dissipative sector. Following the Eckart first-order formalism, the bulk viscosity coefficient is coupled to the instantaneous expansion rate:

$$\zeta_0 = \frac{\mathrm{KAL}_0}{8\pi G}\,H. \tag{11}$$

This delivers a *constant-coupling* viscous pressure $\Pi_0 = -3\zeta_0 H = -3\,\mathrm{KAL}_0 H^2/8\pi G$, which governs the background acceleration (Eq. 23) and fixes the present-day effective equation of state $w_0 = -T_r/M_v$ algebraically (Section 2.2).

Physical motivation for $\zeta \propto H$. Bulk viscosity proportional to the Hubble rate arises naturally in kinetic theory when the dominant relaxation time is set by the expansion itself [Weinberg, 1971, Brevik et al., 2005]. Within SSEE the coupling constant is not fitted but predicted geometrically as $\mathrm{KAL}_0 = (\pi + \varphi)/2 + \pi$.

2.3.2 Scale-factor evolution and the algebraic determination of w_a

The constant-coupling viscosity of the preceding section fixes w_0 algebraically but predicts a *time-independent* effective equation of state. DESI DR2 data require a time-varying component; within SSEE this must arise from a scale-factor-dependent extension of the viscosity coefficient.

Structural requirements on $\zeta(a)$. Any admissible SSEE viscosity extension $\zeta(a)$ must satisfy three independent constraints:

(i) **De Sitter endpoint:** $\zeta(a) \to 0$ as $a \to 1$, so that no viscous term shifts the present-day equation of state away from the k-essence value $w_0 = -T_r/M_v$.

(ii) **Dimensional consistency:** $[\zeta] = [\rho_{\mathrm{DE}}/H]$, constructed from the available dynamical quantities $\{\rho_{\mathrm{DE}}, H, a\}$.

(iii) **Algebraic closure:** the dimensionless amplitude must be a ratio of SSEE structural constants derived exclusively from $\{\varphi, \pi\}$.

The unique admissible form satisfying (i) and (ii) at leading order in $(1-a)$ — the natural expansion around the de Sitter endpoint — is

$$\zeta(a) = \Gamma \, \frac{(1-a)\,\rho_{\mathrm{DE}}(a)}{H(a)}, \tag{12}$$

where Γ is a dimensionless amplitude to be fixed by constraint (iii).

Algebraic determination of Γ. The k-essence sector has already employed the ratio T_r/M_v to fix $|w_0|$. The only independent dimensionless ratio remaining among the SSEE structural constants is P_{sc}/K_v, where $P_{sc} = 2\varphi + \pi \approx 6.378$ (Dynamical Evolution Scalar) and $K_v = 2(\varphi + \pi) \approx 9.519$ (Structural Constraint). The sign is determined by the physical requirement that dark-energy density decreases with time in the observationally favoured phantom-crossing regime:

$$\Gamma = -\frac{P_{sc}}{K_v} = -\frac{2\varphi + \pi}{2(\varphi + \pi)}, \tag{13}$$

yielding the unique SSEE-admissible viscosity:

$$\boxed{\zeta(a) = -\frac{2\varphi + \pi}{2(\varphi + \pi)} \, \frac{(1-a)\,\rho_{\mathrm{DE}}(a)}{H}.} \tag{14}$$

Algebraic consistency with CPL. The CPL parametrisation $w(a) = w_0 + w_a(1-a)$ is the standard model-independent description of slowly-varying dark energy; it serves here as the observational language, not as a physical assumption of SSEE. Adopting CPL and substituting $\zeta(a)$ from Eq. (14) into the SSEE conservation equation (24), one finds algebraic consistency if and only if

$$\boxed{w_a = -\frac{P_{sc}}{K_v} = -\frac{2\varphi + \pi}{2(\varphi + \pi)} \approx -0.6699.} \tag{15}$$

This is not a derived prediction in the sense that CPL is deduced from the action; rather, it establishes that *the SSEE viscosity uniquely fixes the CPL coefficient w_a from the algebraic structure of $\{\varphi, \pi\}$*. The quantity w_a is therefore not a free parameter of the model but a computable consequence of the two-axiom system, in the same sense that w_0 is fixed by the k-essence exponent n. At $a = 1$ the viscous correction vanishes ($\zeta \to 0$), recovering the pure k-essence regime $w = w_0$.

The implied dark-energy density evolution reads:

$$\rho_{\rm DE}(a) = \rho_{\rm DE,0}\, a^{-3(1+w_0+w_a)}\, e^{-3w_a(1-a)}, \tag{16}$$

consistent with the CPL conservation equation

$$\dot{\rho}_{\rm DE} + 3H\,\rho_{\rm DE}\,(1 + w_0) = -3H\,w_a\,(1-a)\,\rho_{\rm DE}, \tag{17}$$

with the effective viscous pressure

$$\Pi(a) = w_a\,(1-a)\,\rho_{\rm DE}(a). \tag{18}$$

At $a \ll 1$ the factor $(1-a) \to 1$ and $\rho_{\rm DE}$ redshifts according to Eq. (16), so ζ remains finite and well-behaved throughout cosmic history.

Boundary conditions. The full viscosity interpolates between two algebraically-fixed regimes:

$$a \to 0 \text{ (early)}: \quad \zeta \to -\frac{2\varphi + \pi}{2(\varphi + \pi)}\,\frac{\rho_{\rm DE,0}\, a^{-3(1+w_0+w_a)}\, e^{-3w_a}}{H(a)}, \tag{19}$$

$$a \to 1 \text{ (today)}: \quad \zeta \to 0. \tag{20}$$

Both limits are free of divergences and require no new parameters.

The viscous pressure is therefore

$$\Pi(a) = \underbrace{-\frac{3\,\mathrm{KAL}_0\,H^2}{8\pi G}}_{\text{background}}\underbrace{-\;3H\,w_a\,(1-a)\,\rho_{\rm DE}(a)}_{\text{CPL evolution}}, \tag{21}$$

where the first term is the constant-coupling Eckart pressure that sustains $\ddot{a} > 0$ and the second term generates the running $w(a)$ observed by DESI DR2 [Abdul-Karim et al., 2025].

2.4 Modified Friedmann Equations

Applying the principle of least action ($\delta S = 0$) to the FLRW metric $ds^2 = -dt^2 + a^2(t)\,d\mathbf{x}^2$ yields three governing equations for the SSEE background.

I. Energy equation:

$$H^2 = \frac{8\pi G}{3}\big(\rho_m + \rho_r + \rho_{\rm SSEE}\big). \tag{22}$$

II. Acceleration equation:

$$\frac{\ddot a}{a} = -\frac{4\pi G}{3}\left[\rho_m + 2\rho_r + \rho_{\rm SSEE}(1+3w_0) - \frac{9\,{\rm KAL}_0\,H^2}{8\pi G}\right]. \tag{23}$$

III. Conservation equation with dissipative source:

$$\dot\rho_{\rm SSEE} + 3H\,\rho_{\rm SSEE}(1+w_0) = \frac{9\,{\rm KAL}_0\,H^3}{8\pi G}. \tag{24}$$

Derivation of the coefficient 9. From Eq. (11), the viscous pressure is $\Pi = -3{\rm KAL}_0 H^2/8\pi G$. In the Raychaudhuri equation the effective pressure of the SSEE sector enters as $p_{\rm eff} = w_0\rho_{\rm SSEE} + \Pi$, so the acceleration term becomes

$$\begin{aligned} -\frac{4\pi G}{3}\big(\rho_{\rm SSEE} + 3p_{\rm eff}\big) &= -\frac{4\pi G}{3}[\rho_{\rm SSEE}(1+3w_0) + 3\Pi] \\ &= -\frac{4\pi G}{3}\left[\rho_{\rm SSEE}(1+3w_0) - \frac{9\,{\rm KAL}_0\,H^2}{8\pi G}\right], \end{aligned} \tag{25}$$

which is exactly Eq. (23). The factor 9 arises from $3 \times |\Pi|/(H^2) = 3 \times 3{\rm KAL}_0/8\pi G$; a factor of 3 rather than 9 would correspond to including Π only once instead of as the trace of the viscous stress.

Similarly, the conservation law $\nabla_\mu T^{\mu\nu} = 0$ for the effective fluid (w_0, Π) gives $\dot\rho + 3H(\rho + w_0\rho + \Pi) = 0$, hence $\dot\rho + 3H\rho(1+w_0) = 9{\rm KAL}_0 H^3/8\pi G$, confirming that Eq. (23) and Eq. (24) are mutually consistent.

2.5 Israel–Stewart Viscous Perturbations

The Eckart formulation used in §2.2–2.4 is a first-order theory that violates causality [Hiscock & Lindblom, 1985]: the viscous signal propagates instantaneously. Israel–Stewart (IS) theory [Israel & Stewart, 1979] resolves this by promoting Π to a dynamical variable with a finite relaxation time τ_Π:

$$\tau_\Pi H_0\,\tilde h(a)\,\frac{{\rm d}\tilde\Pi}{{\rm d}\ln a} + \tilde\Pi = -\,{\rm KAL}_0\,\tilde h^2(a), \tag{26}$$

where $\tilde h = H/H_0$, $\tilde\Pi = \Pi\,8\pi G/H_0^2$, and all densities are normalised to $\rho_{\rm crit,0}$. In the limit $\tau_\Pi \to 0$, Eq. (26) reduces to the Eckart result $\tilde\Pi = -{\rm KAL}_0\tilde h^2$.

Causality constraint. The bulk-viscous sound speed in IS theory is [Hiscock & Lindblom, 1985]

$$c_s^2 = \frac{\zeta}{\rho_{\rm DE}\,\tau_\Pi} = \frac{{\rm KAL}_0}{3\,\Omega_{\rm DE}\,(\tau_\Pi H_0)} \le 1. \tag{27}$$

Setting $c_s^2 = 1$ (the causality boundary) uniquely fixes the IS relaxation time as an algebraic combination of SSEE constants:

$$\tau_\Pi H_0 \;=\; \frac{{\rm KAL}_0}{3\,\Omega_{\rm DE}} \;=\; \frac{{\rm KAL}_0\,M_v}{3\,T_r} \;\approx\; 2.191, \tag{28}$$

with no free parameters.

Growth index. Coupling Eq. (26) to the conservation equation (24) and the linear-growth ODE

$$\delta'' + \left[2 + \frac{\mathrm{d}\ln\tilde{h}}{\mathrm{d}\ln a}\right]\delta' = \frac{3}{2}\,\frac{\Omega_{m,\mathrm{dyn}}\,a^{-3}}{\tilde{h}^2(a)}\,\delta, \tag{29}$$

we solve the system numerically (`src/ssee_is_growth.py`) with the IS background $\tilde{h}^2(a) = \Omega_{m,\mathrm{dyn}}\,a^{-3} + \rho_{\mathrm{DE}}(a)$. The growth rate $f \equiv \mathrm{d}\ln\delta/\mathrm{d}\ln a \approx \Omega_m(a)^\gamma$ is well described by a power law over $0.1 \leq a \leq 1$ with

$$\gamma_{\mathrm{IS}} = 0.657 \pm 0.002 \quad \text{(essentially independent of } \tau_\Pi H_0\text{)}. \tag{30}$$

This differs by 6% from the algebraic sovereignty $\gamma = \varphi^{-1} = 0.618$. Paper 3 §5.4 reports both values: the algebraic prediction $\gamma_{\mathrm{alg}} = 0.618$ and the IS numerical result $\gamma_{\mathrm{IS}} = 0.657$. The IS formalism provides the causal theoretical structure; whether $\gamma = \varphi^{-1}$ is the true attractor of the IS boundary conditions remains a target prediction for forthcoming Stage-IV weak-lensing surveys (LSST, Euclid). Note that the direct IS growth-factor integral (this Appendix) yields $S_8 = 0.837$, while Paper 3's CAMB post-processing method with the same $\gamma_{\mathrm{IS}} = 0.657$ gives $S_8 = 0.818$; both are consistent with Planck 2018 (0.832 ± 0.013) within 1σ.

S_8 constraint. Using the IS growth factor $D_{\mathrm{IS}}(a)$ and the CMB-sector matter density $\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathcal{M} = 0.3199$ (MIRA, Paper 3),

$$S_8^{\mathrm{IS}} \equiv \sigma_{8,\mathrm{IS}}\sqrt{\frac{\Omega_{m,\mathrm{CMB}}}{0.3}} \approx 0.837, \tag{31}$$

within 0.4σ of the Planck-2018 value 0.832 ± 0.013 [Planck Collaboration VI, 2020]. Using the dynamic sector $\Omega_{m,\mathrm{dyn}} = 0.160$ instead gives $S_8 \approx 0.59$ (6σ tension); the correct physical choice depends on the formal resolution of the two-sector Friedmann problem (Section 2.8, item 3).

2.6 Physical Regimes

Late universe ($z \lesssim 2$, DESI DR2). At low redshift $H \approx H_0$, so the viscous source $\propto \mathrm{KAL}_0 H^3/8\pi G$ is small compared to ρ_{SSEE}. The acceleration equation is dominated by $\rho_{\mathrm{SSEE}}(1+3w_0) < 0$ (since $w_0 \approx -0.840 < -1/3$), driving $\ddot{a} > 0$ without dark matter. This regime is the one tested against DESI DR2 in Paper 2, where the algebraic prediction (w_0, w_a) matches the data at 0.05σ.

Early universe (CMB epoch, $z \sim 10^3$). At recombination $H \gg H_0$, the geometric retention term $9\,\mathrm{KAL}_0 H^3/8\pi G$ acts as a dissipative source in Eq. (24), injecting energy into the dark-energy fluid and modifying its effective equation of state. This structural viscosity provides the physical mechanism by which an effective matter density $\Omega_{m,\mathrm{eff}} = 1 - T_r/M_v \approx 0.160$ yields the correct CMB acoustic scale through the MIRA factor $\mathcal{M} = 2 - 1/\varphi^2 \approx 1.9989$ (Paper 3, §2.3).

2.7 Inflationary Embedding: α-Attractor with $\alpha = \varphi^4/3$

The SSEE algebraic predictions for the primordial scalar spectral index $n_s = 1 - \varphi^{-7}$ (Paper 1, §1.1) and the tensor-to-scalar ratio $r = \varphi^{-10}$ (Paper 4) belong simultaneously to a well-defined class of inflationary models: the *α-attractors* of Kallosh & Linde [2013].

α-attractor predictions. In the universal leading-order limit, α-attractor models predict

$$n_s = 1 - \frac{2}{N}, \qquad r = \frac{12\alpha}{N^2}, \tag{32}$$

where N is the number of inflationary e-folds and $\alpha > 0$ parameterises the inverse curvature of the scalar field's Kähler target manifold [Kallosh & Linde, 2013, Ferrara et al., 2013]. Setting $\alpha = 1$ recovers Starobinsky R^2 inflation exactly.

SSEE identification. Inserting $n_s = 1 - \varphi^{-7}$ into the first relation gives

$$N = 2/\varphi^{-7} = 2\varphi^7 \approx 58.07, \tag{33}$$

which lies squarely in the observationally required window $50 \lesssim N \lesssim 60$. Substituting $N = 2\varphi^7$ and $r = \varphi^{-10}$ into the second relation:

$$\alpha = \frac{r\,N^2}{12} = \frac{\varphi^{-10} \cdot 4\varphi^{14}}{12} = \frac{4\varphi^4}{12} = \frac{\varphi^4}{3}. \tag{34}$$

This result is algebraically exact (verified numerically to machine precision; see `src/ssee_inflation_co`
Using the Fibonacci identity $\varphi^4 = 3\varphi + 2$:

$$\alpha = \frac{3\varphi + 2}{3} = \varphi + \frac{2}{3} \approx 2.285. \tag{35}$$

Geometric interpretation. In the α-attractor framework the Kähler curvature of the inflaton's target space (a Poincaré disk of radius $\sqrt{3\alpha}$) is $\mathcal{R} = -2/(3\alpha)$. For $\alpha = \varphi^4/3$:

$$\mathcal{R} = -\frac{2}{3 \cdot \varphi^4/3} = -\frac{2}{\varphi^4} = -\varphi^{-4} \approx -0.146. \tag{36}$$

The inflaton therefore lives on a hyperbolic manifold whose curvature is set by φ alone. This gives a *geometric* origin for the appearance of φ in inflationary observables: φ enters not as a numerical coincidence but as the intrinsic length scale of the Kähler geometry of the SSEE inflaton sector.

Slow-roll parameters. The SSEE predictions $(n_s, r) = (1-\varphi^{-7}, \varphi^{-10})$ fix the slow-roll parameters uniquely:

$$\varepsilon = \frac{r}{16} = \frac{\varphi^{-10}}{16}, \tag{37}$$

$$\eta = \frac{n_s - 1 + 6\varepsilon}{2} = \frac{-\varphi^{-7} + 3\varphi^{-10}/8}{2}, \tag{38}$$

$$\frac{\eta}{\varepsilon} = 3 - 8\varphi^3 = -5 - 16\varphi \approx -30.9. \tag{39}$$

The ratio $\eta/\varepsilon = 3 - 8\varphi^3$ is a pure algebraic identity derived from the SSEE axioms $\{\varphi, \pi\}$; it defines a one-parameter family of inflationary potentials $V(\phi_{\text{inf}})$ whose single element consistent with α-attractors is $\alpha = \varphi^4/3$.

Relation to Starobinsky. Starobinsky ($\alpha = 1$) reproduces n_s correctly for $N = 2\varphi^7$ but gives $r_{\text{Staro}} = 3/\varphi^{14} \approx 0.00356$, a factor of $\varphi^4/3 \approx 2.28$ below the SSEE prediction φ^{-10}. SSEE is therefore *not* Starobinsky inflation; it is a generalised α-attractor that reduces to Starobinsky only in the limit $\varphi \to 1$ (i.e. $\alpha \to 1$, the fixed point of the golden-ratio recursion $x = 1 + 1/x$).

2.8 Summary and Open Problems

Table 2 summarises the EFT status. The action (5) with kinetic exponent (8) and viscosity (14) constitutes a *complete*, zero-free-parameter EFT description of the SSEE background evolution. Every constant (n, KAL_0, w_0, w_a, Ω_{DE}) is derived from φ and π alone; only the overall energy scale H_0 enters as an observational input. Notably, $w_a = -P_{sc}/K_v$ is *not* postulated but emerges as the unique algebraic amplitude of the viscosity's scale-factor evolution (Section 2.3.2).

Four aspects require further development:

1. **Growth index.** The IS-derived growth index $\gamma_{\mathrm{IS}} = 0.657$ (§2.5) differs by 6% from the algebraic prediction $\varphi^{-1} = 0.618$. A mechanism that would select $\gamma = \varphi^{-1}$ from IS boundary conditions is not yet identified; this constitutes a near-prediction testable by LSST/Euclid Stage-IV weak lensing.

2. **Full perturbation spectrum.** The IS growth ODE (29) governs sub-Hubble dark-matter perturbations; the matching to CMB scalar transfer functions and the tensor-mode propagation in the IS background require a complete Boltzmann implementation.

3. **Radiative stability.** The EFT cutoff scale Λ_{UV} and the quantum stability of the $n < 0$ power-law sector are not yet established.

4. **Inflation–dark energy unification.** The α-attractor sector (§2.7) and the k-essence dark-energy sector (§2.2) both originate from φ, suggesting a single unified action. Constructing the quintessential inflation potential $V(\phi)$ that reduces to $P(X) = \mathcal{A}X^n$ at late times and to an $\alpha = \varphi^4/3$ attractor at early times is identified as high-priority future work.

Table 2: EFT status summary for SSEE-V3.6.

Element	Status	Derivation
Action form S	✓Complete	Standard GR + k-essence + Eckart
$u_\mu(\phi)$ coupling	✓Complete	$u_\mu = -\partial_\mu\phi/\sqrt{2X}$
Kinetic exponent n	✓Algebraic	$n = (T_r - M_v)/2T_r$ from φ, π
Equation of state w_0	✓Algebraic	$w_0 = 1/(2n-1) = -T_r/M_v$
Viscosity coupling ζ	✓Algebraic	$\zeta = \mathrm{KAL}_0 H/8\pi G$
Friedmann eqs. I–III	✓Consistent	Factor 9 verified in §2.4
w_a from EFT	✓Algebraic	$\zeta(a) \propto (1-a)\rho_{\mathrm{DE}}/H$; see §2.3.2
IS relaxation time τ_Π	✓Algebraic	$\tau_\Pi H_0 = \mathrm{KAL}_0/(3\Omega_{\mathrm{DE}}) \approx 2.191$
IS causality ($c_s^2 \leq 1$)	✓Verified	Eq. (27); $c_s^2 = 1$ at boundary
Growth index γ_{IS}	✓Numerical	0.657 ± 0.002 (6% above $\varphi^{-1} = 0.618$)
S_8 (CMB sector)	✓Numerical	0.837, within 0.4σ of Planck
α-attractor ($\alpha = \varphi^4/3$)	✓Algebraic	Eq. (34); exact, $N = 2\varphi^7$
Kähler curvature $\mathcal{R} = -\varphi^{-4}$	✓Algebraic	Eq. (36)
Full perturbation spectrum	∘ Open	Off-diagonal CMB cov. + Stage-IV WL
Inflation–DE unification	∘ Open	Quintessential inflation potential
Radiative stability	∘ Open	Future work

3 Dark Energy Evolution

SSEE models dynamical dark energy via the Chevallier–Polarski–Linder (CPL) form $w(a) = w_0 + w_a(1-a)$. Motivated by Eq. (4), the expansion is subject to an effective geometric pressure:

$$H^2 = \frac{8\pi G}{3}(\rho_{\rm bar} + \rho_{\rm DE}), \qquad \rho_{\rm DE} = \rho_{\rm crit,0}\left(\frac{T_r}{M_v}\right). \tag{40}$$

The baseline state w_0 is set by the static geometric ratio T_r/M_v, and the evolution w_a encodes the dynamical departure from saturation driven by P/K_v:

$$w_0 = -\frac{T_r}{M_v} \approx -0.840, \qquad w_a = -\frac{P}{K_v} \approx -0.670. \tag{41}$$

Explicit reduction to the two axioms. Substituting the algebraic definitions $T_r = \frac{3(3\varphi+\pi)}{2}$ (TRIAL), $M_v = 3(\varphi+\pi)$ (MIKAEL_V), $P_{sc} = 2\varphi+\pi$ (PYROS), and $K_v = 2(\varphi+\pi)$ (KRYSTOS), both dark-energy observables collapse to closed forms in φ and π alone:

$$\boxed{w_0 = -\frac{3\varphi+\pi}{2(\varphi+\pi)} = -0.8399}, \qquad \boxed{w_a = -\frac{2\varphi+\pi}{2(\varphi+\pi)} = -0.6699}. \tag{42}$$

No fitting procedure is involved: these values are uniquely fixed by the geometry of φ and π. The numerators $3\varphi+\pi$ and $2\varphi+\pi$ reflect the three- and two-fold contributions of φ to the saturation horizon, while the common denominator $2(\varphi+\pi)$ is twice the Stability Metric $\Omega = \varphi+\pi$.

Asymptotic saturation. Although $w_0 + w_a = -1.510$ would ordinarily imply a phantom Big Rip, SSEE adopts CPL only as a local observational parameterisation. In the far future $(a \to \infty)$ the energy density saturates at the finite T_r/M_v boundary of Eq. (4), ensuring universal stability.

DESI DR2 [Abdul-Karim et al., 2025] analyses indicate an evolving preference for dynamical dark energy. The SSEE algebraic point $(w_0, w_a) = (-0.840, -0.670)$ falls within the 68% confidence contours of the combined BAO+CMB+DESY5 datasets (Mahalanobis $\chi^2_{2D} = 0.080$, i.e. 0.05σ; quantified in Paper 2 [Almeida Vallejo, 2026]).

4 Galaxy Cluster Dynamics

4.1 The Cluster Mass Discrepancy and IGIMF

In rich galaxy clusters, inferred dynamical mass typically exceeds the visible baryonic mass by a factor of ∼5–6. Recent literature incorporating a variable IGIMF [Zhang et al., 2026] demonstrates that baryonic masses have been systematically underestimated: heavy stellar remnants raise the effective baryonic baseline to at least $88^{+5}_{-4}\%$ of the MOND dynamical mass. In SSEE, this updated IGIMF baseline multiplied by $\mathrm{KAL}_0 \approx 5.5214$ natively closes the remaining discrepancy without cold dark matter.

4.2 Structural Viscosity and the Neutrino Bridge

At the cluster scale, the total effective dynamical mass $M_{\rm dyn}$ within radius R is:

$$M_{\rm dyn}(R) = M_{\rm bar}^{\rm IGIMF}(R) \cdot \mathrm{KAL}_0 \cdot [1 + f_\nu], \tag{43}$$

where f_ν quantifies the fractional mass contribution of trapped relic neutrinos. Using $\sum m_\nu = 0.085\,\mathrm{eV}$ — robustly permitted within SSEE's dynamic dark-energy regime — the cosmological neutrino density is $\rho_\nu \approx 28.6\,\mathrm{eV\,cm^{-3}}$. For a cluster with escape velocity $v_{\mathrm{esc}} \approx 2000\,\mathrm{km\,s^{-1}}$, integration of the Fermi–Dirac distribution [Tremaine & Gunn, 1979] yields a trapped fraction $f_\nu \approx 0.018$–0.022. This structurally consistent $\sim 2\%$ correction provides the final macroscopic bridge.

4.3 Quantitative Cluster Analysis

Applying Eq. (43) to four well-documented clusters demonstrates predictive accuracy against observed dynamical masses (Table 3).

Table 3: Cluster mass predictions. All masses within R_{200} in units of $10^{14}\,M_\odot$. $M_{\mathrm{SSEE}} = M_{\mathrm{bar}}^{\mathrm{IGIMF}} \times 5.5214 \times 1.02$. IGIMF baselines from Zhang et al. [2026]; observed masses from Simionescu et al. 2011, Walker et al. 2012, Pointecouteau et al. 2004, Clowe et al. 2006.

Cluster	$M_{\mathrm{bar}}^{\mathrm{IGIMF}}$ $[10^{14}\,M_\odot]$	M_{SSEE} $[10^{14}\,M_\odot]$	$M_{\mathrm{dyn}}^{\mathrm{obs}}$ $[10^{14}\,M_\odot]$
Coma	1.8 ± 0.2	10.1 ± 1.1	9.8 ± 1.0
A2029	2.2 ± 0.2	12.4 ± 1.1	12.0 ± 1.2
A478	1.5 ± 0.2	8.4 ± 1.1	8.0 ± 1.0
Bullet (main)	1.2 ± 0.2	6.7 ± 1.1	6.5 ± 1.0

All four predictions agree with observed masses at the 1σ level ($\chi_r^2 = 0.122$; full analysis in Paper 2 [Almeida Vallejo, 2026]).

4.4 Gravitational Lensing and the Newtonian Limit

The Bullet Cluster (1E 0657-56) is frequently cited as proof of particulate dark matter due to the spatial offset between collisional X-ray gas and lensing-mass peaks [Clowe et al., 2006]. SSEE reproduces this topological offset through macroscopic structural viscosity without collisionless particles.

The structural viscosity acts as an environment-dependent scalar field. Defining $x = |\nabla\Phi|/a_0$, SSEE adopts an interpolation function analogous to the canonical MOND simple interpolating function $\mu(x)$:

$$\mu_{\mathrm{SSEE}}(x) = \frac{x+1}{x+\mathrm{KAL}_0} \implies \mathrm{KAL}(x) = \frac{1}{\mu_{\mathrm{SSEE}}(x)} = \frac{x+\mathrm{KAL}_0}{x+1}. \tag{44}$$

For $x \gg 1$ (Newtonian limit), $\mathrm{KAL}(x) \to 1$. For $x \ll 1$ (deep MONDian limit), $\mathrm{KAL}(x) \to \mathrm{KAL}_0$. During a high-velocity cluster collision, the stripped X-ray gas flattens its local potential gradient ($x \ll 1$), activating the full KAL_0 amplification; collisionless galaxies retain compact potential wells ($x \geq 1$), coupling non-linearly.

The effective lensing convergence is:

$$\kappa(\vec{\theta}) = \frac{\Sigma_{\mathrm{SSEE}}(\vec{\theta})}{\Sigma_{\mathrm{crit}}}, \quad \Sigma_{\mathrm{SSEE}}(\vec{\theta}) = \int \rho_{\mathrm{bar}}^{\mathrm{IGIMF}}(\vec{\theta},\ell)\cdot \mathrm{KAL}(x)\,d\ell, \quad \Sigma_{\mathrm{crit}} = \frac{c^2}{4\pi G}\frac{D_S}{D_L D_{LS}}. \tag{45}$$

This allows lensing peaks to track galaxies, consistent with QUMOND simulations [Angus et al., 2007].

Data Availability

All observational data used in this work are publicly available. DESI DR2 BAO measurements are from Abdul-Karim et al. [2025] (https://arxiv.org/abs/2503.14738). Galaxy-cluster dynamical masses are from Zhang et al. [2026] (https://arxiv.org/abs/2602.06082). Analysis scripts and data files are available at https://github.com/mikealmeida1721/SSEE.

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References

Abdul-Karim, M., et al. (DESI Collaboration) (2025). DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints. *Phys. Rev. D* **112**, 083515. arXiv:2503.14738.

Almeida Vallejo, M. E. (2026). Bayesian MCMC Validation of the Structural Self-Energy Expansion Framework (SSEE-V3.6). *Paper 2, this series.*

Angus, G. W., et al. (2007). On the proof of dark matter, the law of gravity and the mass of neutrinos. *MNRAS* **374**, 1248.

Clowe, D., et al. (2006). A direct empirical proof of the existence of dark matter. *ApJL* **648**, L109.

Pointecouteau, E., et al. (2004). XMM-Newton observations of the most X-ray luminous galaxy cluster RX J1347.5-1145. *A&A* **423**, 33.

Simionescu, A., et al. (2011). Baryons at the edge of the X-ray brightest galaxy cluster. *Science* **331**, 1576.

Tremaine, S., & Gunn, J. E. (1979). Dynamical role of light neutral leptons in cosmology. *Phys. Rev. Lett.* **42**, 407.

Walker, S. A., et al. (2012). Suzaku observations of the outskirts of A2029. *MNRAS* **422**, 3503.

Zhang, D., Zonoozi, A. H., & Kroupa, P. (2026). Revisiting the missing mass problem in MOND for nearby galaxy clusters. *Phys. Rev. D* **113**, 043027. arXiv:2602.06082.

Eckart, C. (1940). The thermodynamics of irreversible processes. III. Relativistic theory of the simple fluid. *Phys. Rev.* **58**, 919.

Weinberg, S. (1971). Entropy generation and the survival of protogalaxies in an expanding universe. *ApJ* **168**, 175.

Israel, W., & Stewart, J. M. (1979). Transient relativistic thermodynamics and kinetic theory. *Ann. Phys.* **118**, 341.

Brevik, I., Nojiri, S., Odintsov, S. D., & Vanzo, L. (2005). Entropy and universality of the Cardy-Verlinde formula in dark energy universe. *Phys. Rev. D* **70**, 043520.

Hiscock, W. A., & Lindblom, L. (1985). Generic instabilities in first-order dissipative relativistic fluid theories. *Phys. Rev. D* **31**, 725.

Planck Collaboration (Aghanim, N., et al.) (2020). Planck 2018 results VI: Cosmological parameters. *A&A* **641**, A6. arXiv:1807.06209.

Kallosh, R., & Linde, A. (2013). Universality class in conformal inflation. *JCAP* **07**, 002. arXiv:1306.5220.

Ferrara, S., Kallosh, R., Linde, A., & Porrati, M. (2013). Minimal supergravity models of inflation. *Phys. Rev. D* **88**, 085038. arXiv:1307.7696.